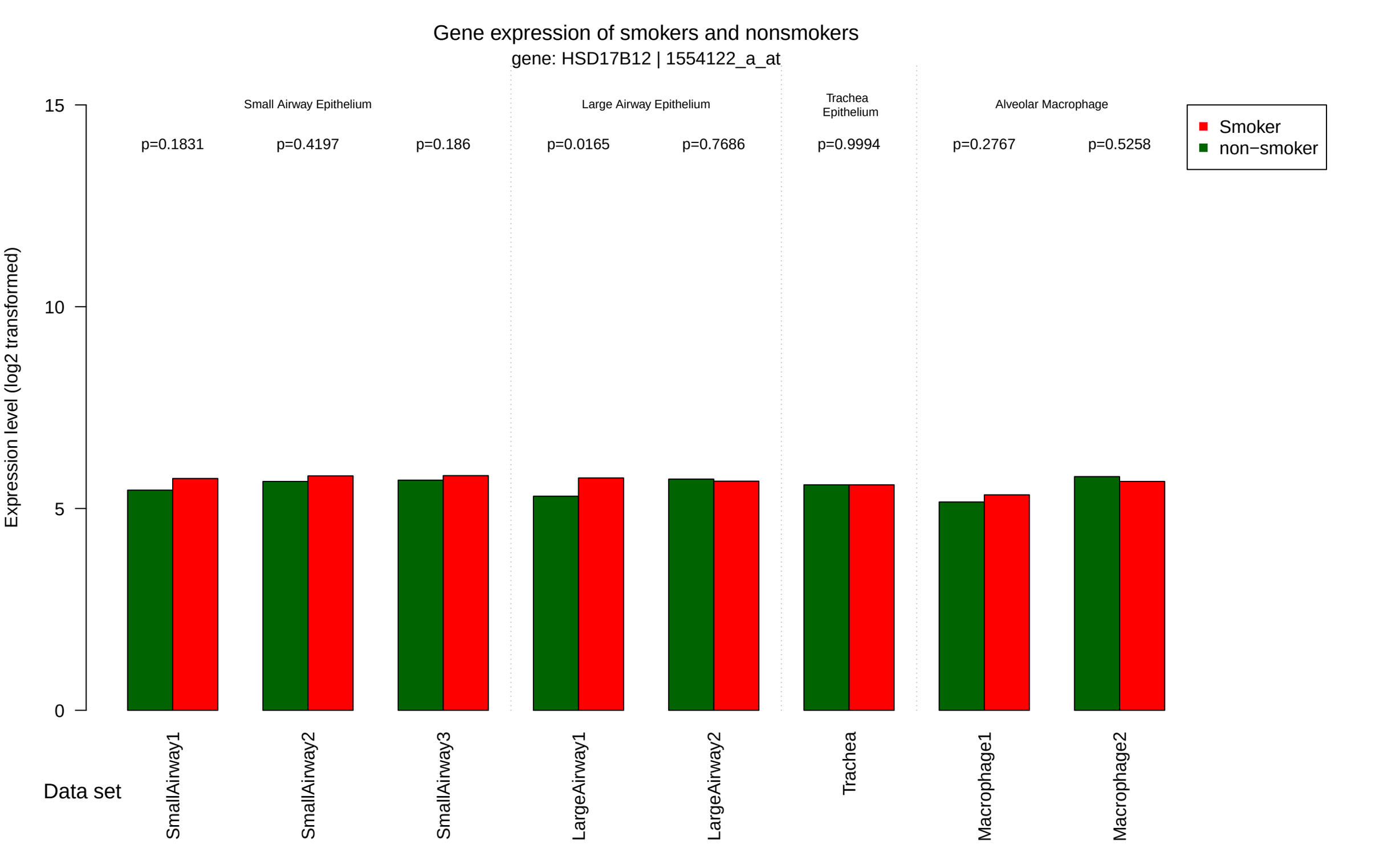
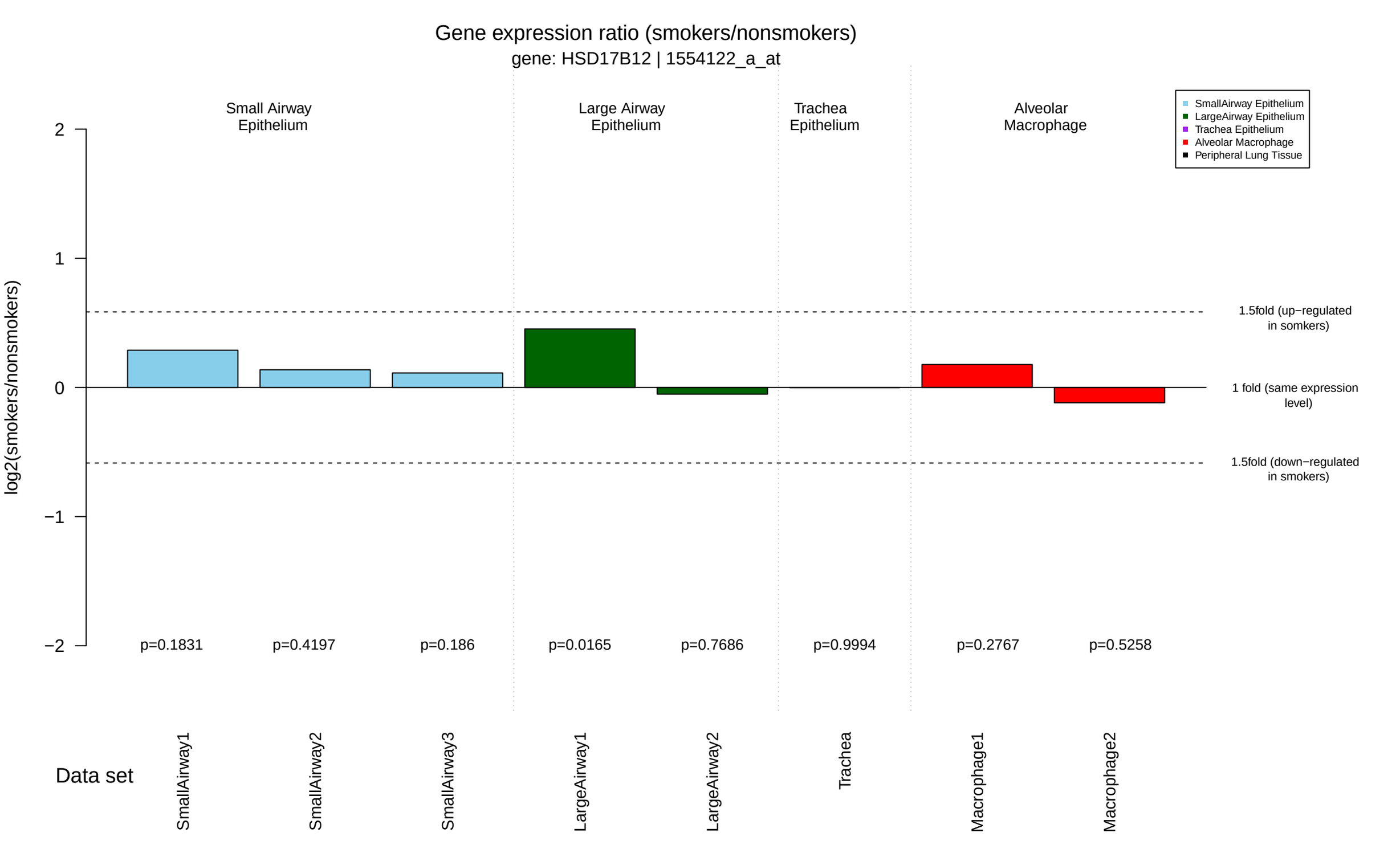
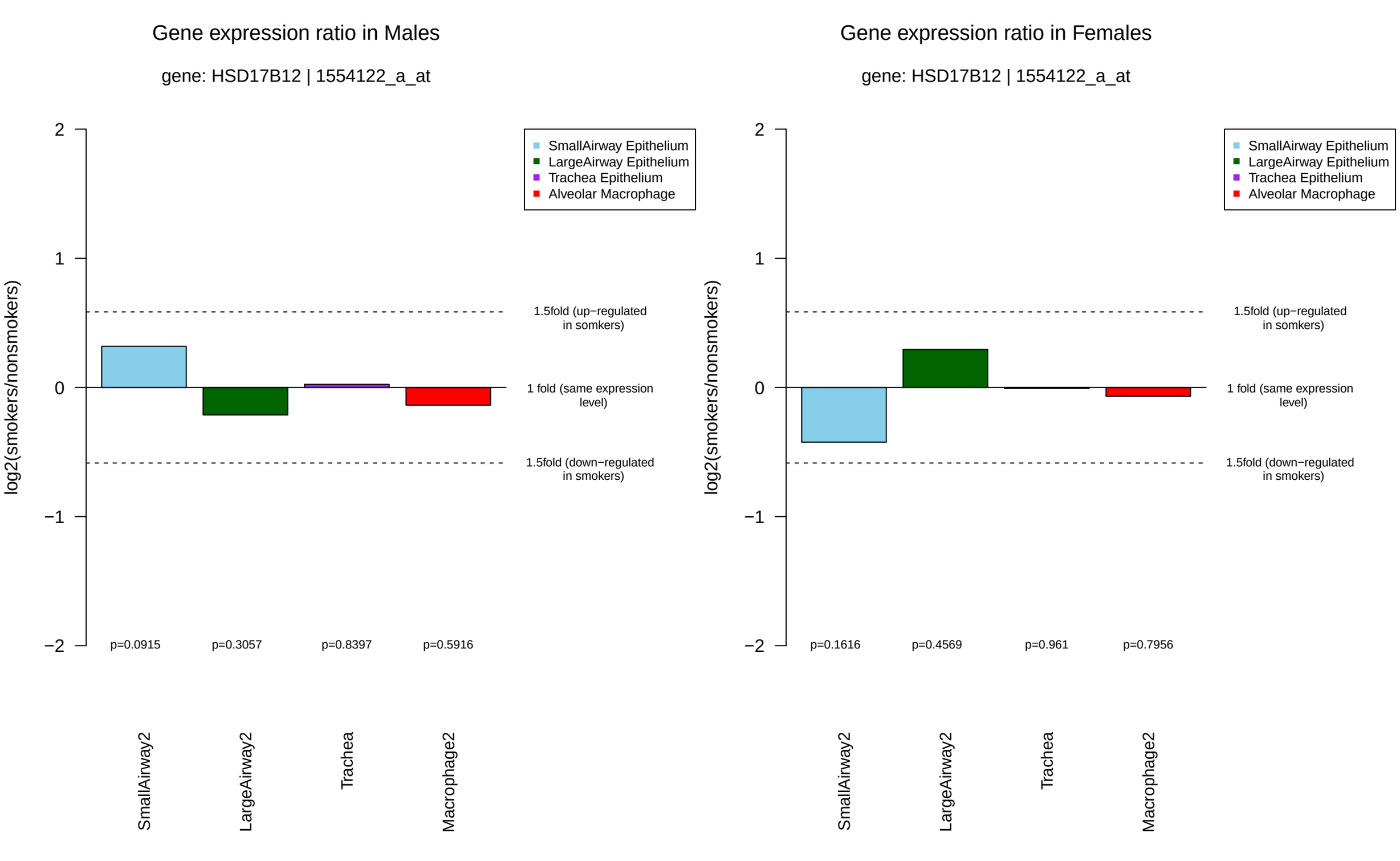
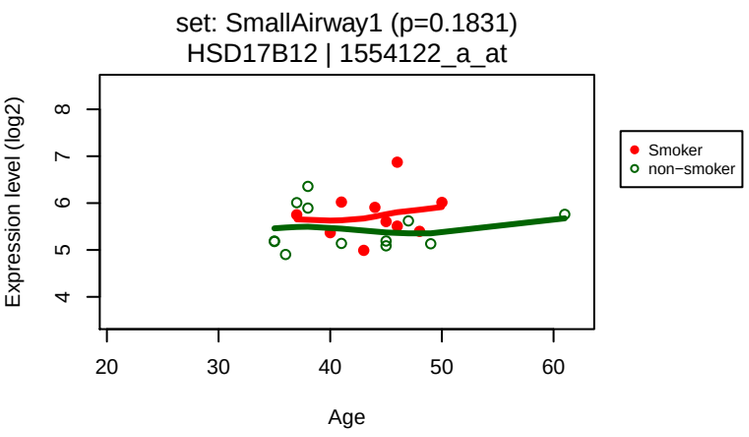
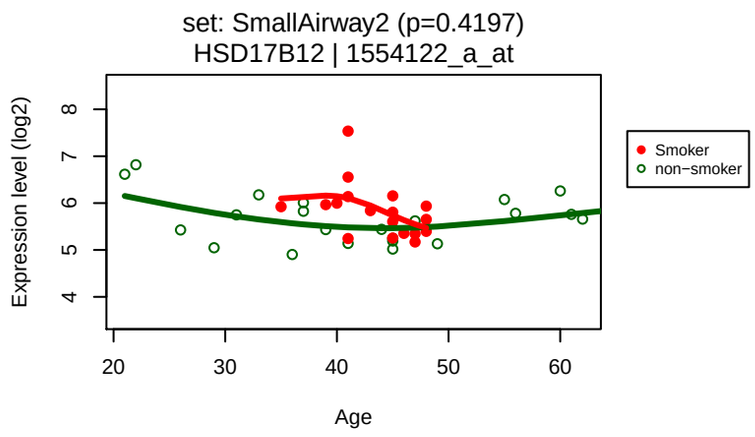
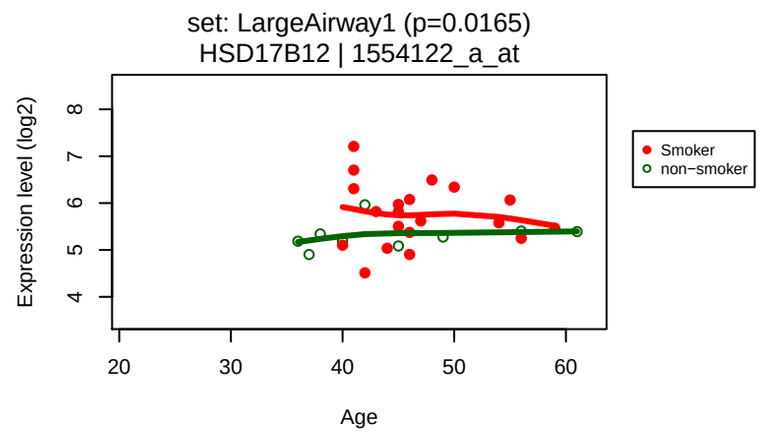

Gene expression of smokers and nonsmokers
gene: HSD17B12 | 1554122_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1831
p=0.4197
p=0.186
p=0.0165
p=0.7686
p=0.9994
p=0.2767
p=0.5258
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

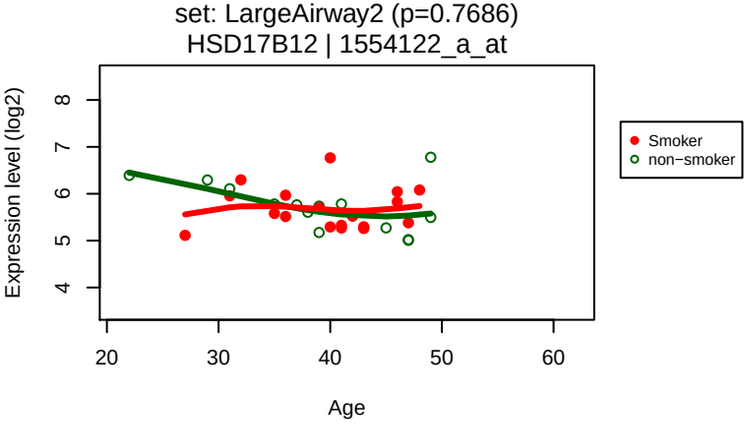

Gene expression ratio (smokers/nonsmokers)
gene: HSD17B12 | 1554122_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1831
p=0.4197
p=0.186
p=0.0165
p=0.7686
p=0.9994
p=0.2767
p=0.5258
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

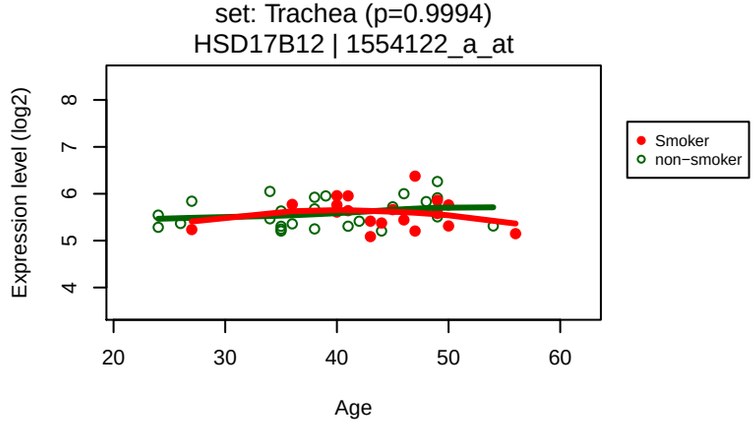

Gene expression ratio in Males
gene: HSD17B12 | 1554122_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0915
p=0.3057
p=0.8397
p=0.5916
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HSD17B12 | 1554122_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1616
p=0.4569
p=0.961
p=0.7956
SmallAirway2
LargeAirway2
Trachea
Macrophage2

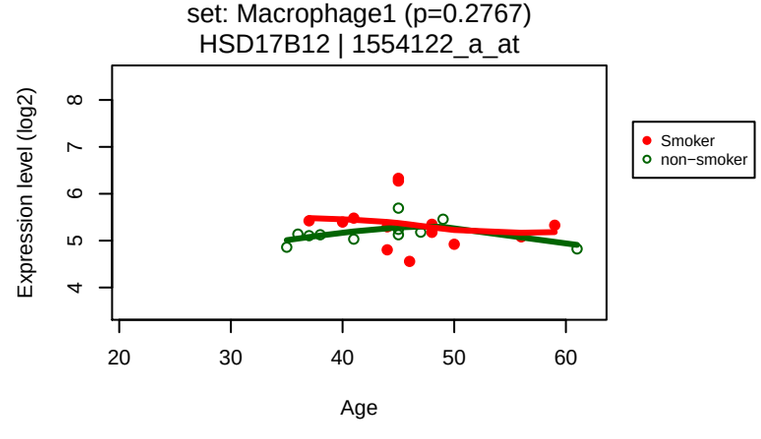

set: SmallAirway1 (p=0.1831)
HSD17B12 | 1554122_a_at
Expression level (log2)
Age
Smoker
non−smoker

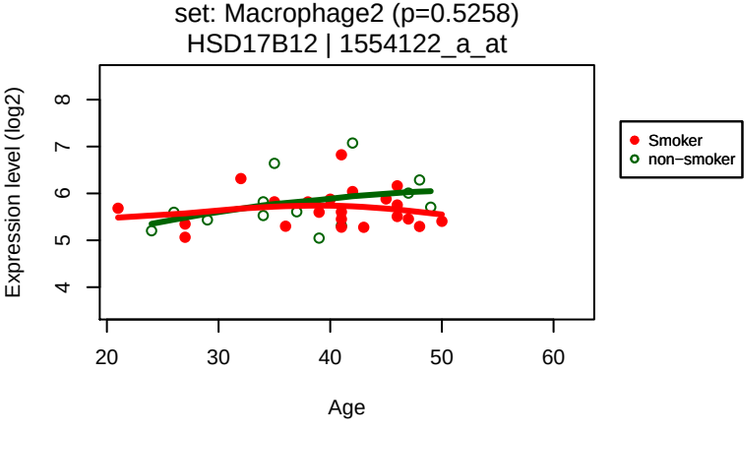

set: SmallAirway2 (p=0.4197)
HSD17B12 | 1554122_a_at
Expression level (log2)
Age
Smoker
non−smoker

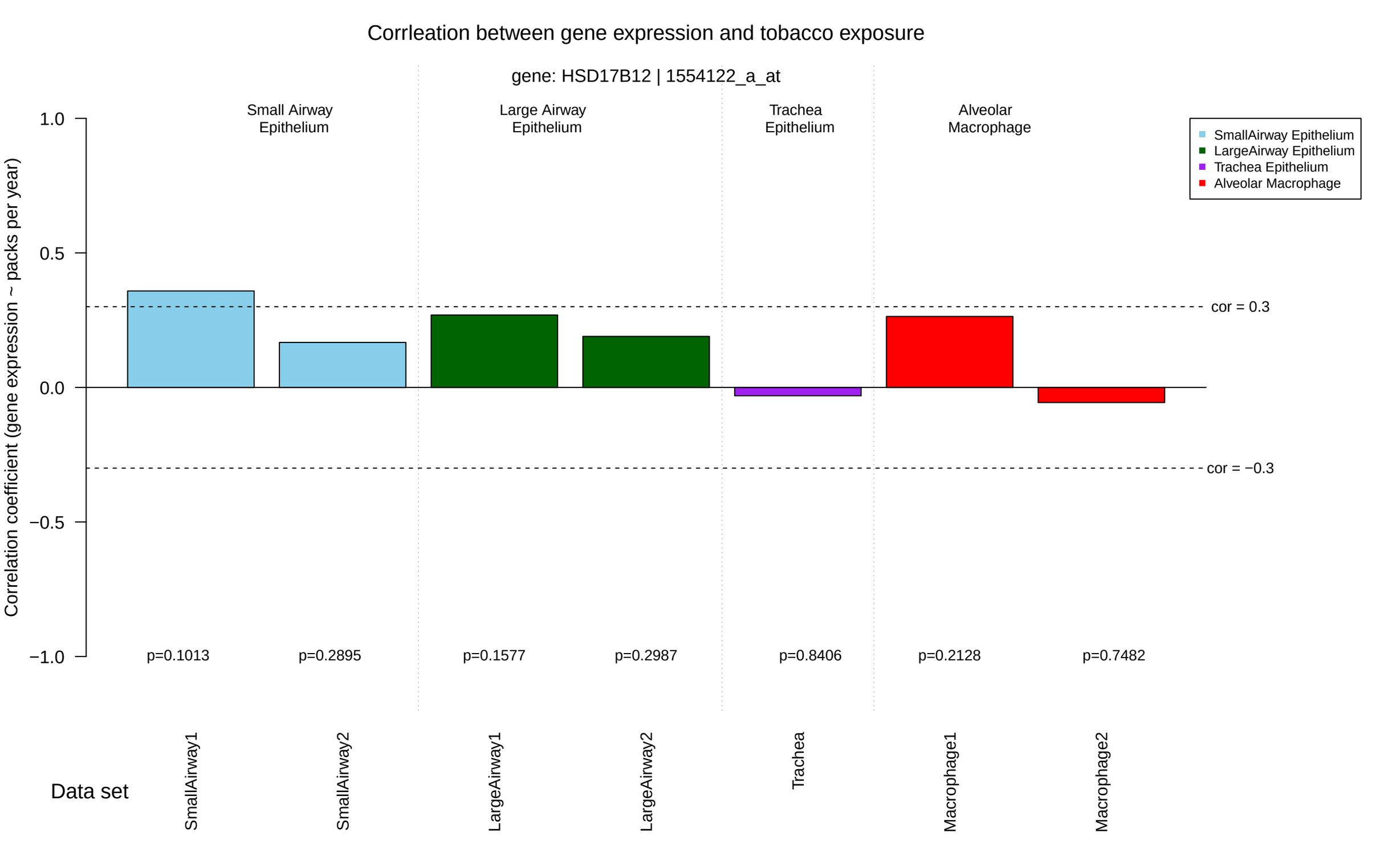

set: LargeAirway1 (p=0.0165)
HSD17B12 | 1554122_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7686)
HSD17B12 | 1554122_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9994)
HSD17B12 | 1554122_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2767)
HSD17B12 | 1554122_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5258)
HSD17B12 | 1554122_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HSD17B12 | 1554122_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1013
p=0.2895
p=0.1577
p=0.2987
p=0.8406
p=0.2128
p=0.7482
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2